Real Estate Price Prediction

https://github.com/aiyshwary/real-estate-price-prediction

Python pandas NumPy scikit-learn Matplotlib Seaborn

OVERVIEW

A supervised regression case study that predicts house prices using the kc_house_data dataset, benchmarking multiple models and comparing performance with R2 and MSE metrics.

IMPACT

Benchmarked six regression models with bagging/boosting and hyperparameter tuning to identify the most accurate predictor for house prices.

TECHNICAL WALKTHROUGH

Project Contents

The project includes a Jupyter notebook, the kc_house_data.csv dataset, and a slide deck summarizing findings.

i) Notebook: ML_HOUSE_RENT_PREDICTION_CASE_STUDY IPYNB (1).ipynb

ii) Dataset: kc_house_data.csv (house attributes and prices)

iii) Slides: ML GROUP 8 (1).pptx

Data Preparation

Loaded the dataset, checked distributions and missing values, and prepared features for regression modeling.

i) Exploratory analysis with pandas, matplotlib, and seaborn.

ii) Basic cleaning to ensure consistent numeric inputs.

Models Evaluated

Benchmarked multiple regression models to compare predictive performance.

i) Linear Regression, KNN Regressor, Decision Tree Regressor

ii) Random Forest, AdaBoost, Gradient Boosting Regressors

Training & Tuning

Applied bagging and boosting techniques, then used hyperparameter tuning to improve generalization.

> i) Model selection guided by R2 and MSE metrics.

Results

Compared model scores and selected the best-performing regressor based on error metrics.

Reproducibility

Notebook can be run end-to-end with the dataset placed in the project root, producing charts and model metrics.

KEY DETAILS

1. Loaded and explored kc_house_data.csv, performing basic cleaning and feature analysis for regression modeling.
2. Trained Linear Regression, KNN Regressor, Decision Tree, Random Forest, AdaBoost, and Gradient Boosting models.
3. Compared model performance using R^2 and MSE to select the best-performing regressor.
4. Applied bagging/boosting strategies and hyperparameter tuning to improve predictive accuracy.
5. Documented the workflow in a Jupyter Notebook and included presentation slides for project reporting.
6. Explicitly excludes logistic regression from the model comparison, noting it is a classification algorithm unsuitable for continuous price prediction.
7. Includes a PowerPoint slide deck alongside the notebook for presentation-ready project documentation.